Dylan Becchetti

Mechanical Engineering | Manufacturing Automation & Engineering Software

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Portfolio: formsby-systems.github.io/engineering-portfolio

ENGINEERING EXPERIENCE

Cardinal Glass Industries | St. Louis Park, MN

Continuous Improvement Engineering Intern, summers 2024–2025 | Engineering Intern, summer 2023

- Built an Excel/VBA tracker for R&D shift debriefs and automated weekly quality and delivery reporting from unit inspections, production exports and work orders.
- Linked purchase-order history to a legacy inventory in Smartsheet, reconciled inconsistent part identifiers, and surfaced items for maintenance review.
- Wrote illustrated assembly instructions for customer samples and planned spare-parts storage and rack signage across 14 locations.
- Measured cleanroom airflow, analyzed pilot-line data in Minitab, drafted AutoCAD equipment layouts and contributed to design failure mode and effects analysis.

SELECTED PROJECTS

Wilbur Powerhouse | Mechanical Design, year-long team capstone

- Built the full-scale OpenSCAD assembly. Widened the ring gear to 79.5 inches and moved the drive while preserving the inner tooth profile.
- Worked with the team on carrier FEA and CNC fabrication. Identified a bore clearance change during fit checks on the first welded gear unit.

Midas Futures | Creator & Technical Lead

- Built batch and ablation workflows that record inputs and checkpoints. In a separate Railway run, reproduced 11 local decision files byte for byte.
- Unified historical and forward rules in Python. Compared one strategy against a frozen reference and matched 228 session decisions and 225 modeled trade lifecycles.

Operalith | Founder & Product Lead, independent product

- Built a React/Next.js workspace where users select tools and features while keeping tasks, projects and inventory records connected.
- Designed PostgreSQL saves that return the original result on retry and reject stale edits. Verified persistence after reload and history-preserving undo in a hosted beta.

Formsby | Creator, housing search prototype

- Built a Next.js and PostgreSQL housing search with an Inngest job queue, deduplication and ranking. Kept results tied to the signed-in user’s latest run and separated failed jobs from empty searches.

EDUCATION

Lehigh University | B.S. Mechanical Engineering, May 2026

Materials Science and Engineering minor expected December 2026. | SOLIDWORKS Associate (CSWA), 2024.

TECHNICAL SKILLS

Python, JavaScript, React/Next.js, SQL/PostgreSQL, Git, automated testing, OpenSCAD, SOLIDWORKS, AutoCAD, Excel/VBA, Smartsheet, Minitab.